



# JKBOSE 2026

## CLASS 11<sup>TH</sup>

### Relations and Functions

**MATHS**

**LECTURE- 04**

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# TOPICS TO BE COVERED

1

PROBLEM DISCUSSION.

2

functions: \* special Type of Relation

3

\* {Real valued function}

\* Every element of A has a unique in B.

\* Input is a Real number

\* output is also a Real number.

\*  $f(x) = \sqrt{x}$  (Square root function)

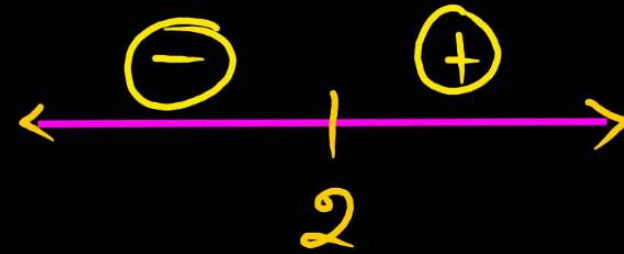
$x \geq 0$  (Defined)

Domain =  $[0, \infty)$

\* Wavy - Curve Method:

\*  $x - 2$

Q) For what values of  $x$ ,  $x - 2 \geq 0$   
 $\& x - 2 \leq 0$ ?



\* To the left of 2,  $x - 2 < 0$

\* To the right of 2,  $x - 2 > 0$

$$\begin{aligned} * \quad x^2 + 5x + 6 &= x^2 + 2x + 3x + 6 \\ &= x(x + 2) + 3(x + 2) \\ &= (x + 2)(x + 3) \end{aligned}$$



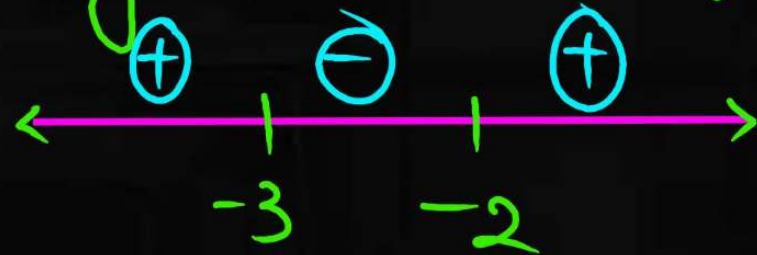
## Example

Find the Domain of the function  $f(x) = \sqrt{x^2 + 5x + 6}$

(Sol) We know that  $f(x) = \sqrt{x^2 + 5x + 6}$  is defined only if  $x^2 + 5x + 6 \geq 0$

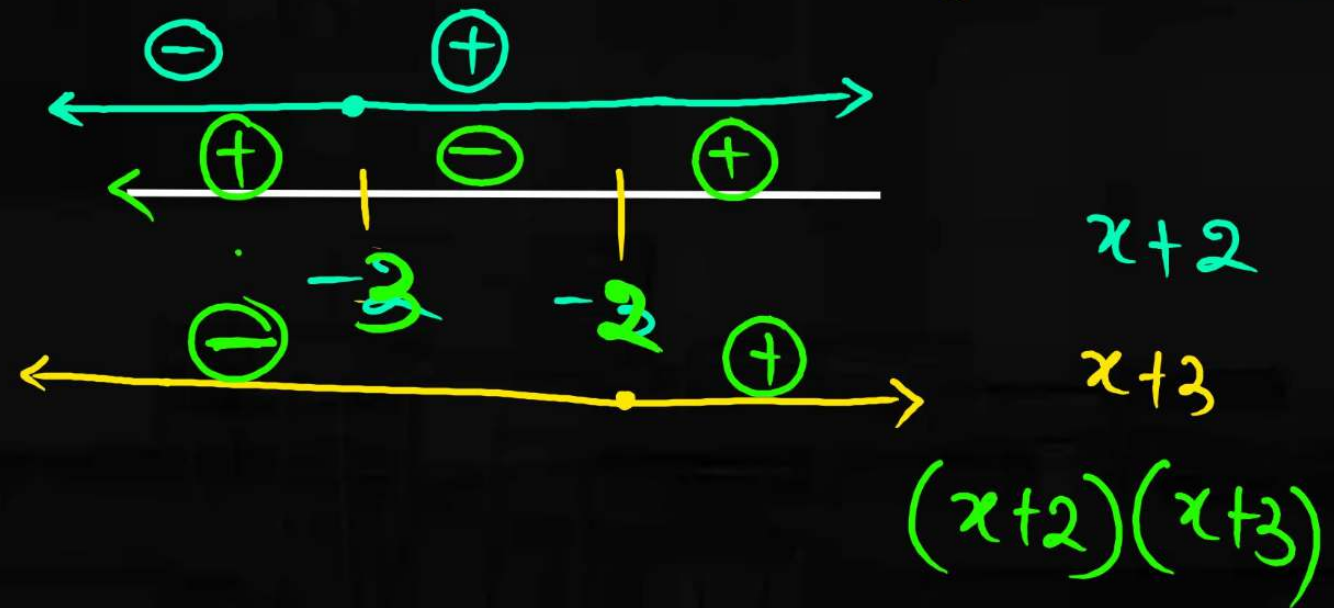
$$\Rightarrow (x+2)(x+3) \geq 0$$

By wavy-curve method; we get



$$\Rightarrow x \in (-\infty, -3] \cup [-2, \infty)$$

Therefore Domain =  $(-\infty, -3] \cup [-2, \infty)$



## Example

Find the Domain and range of the function  $f(x) = \sqrt{25 - x^2}$

(Sol) We know that  $f(x) = \sqrt{25 - x^2}$  is defined only if  $25 - x^2 \geq 0$

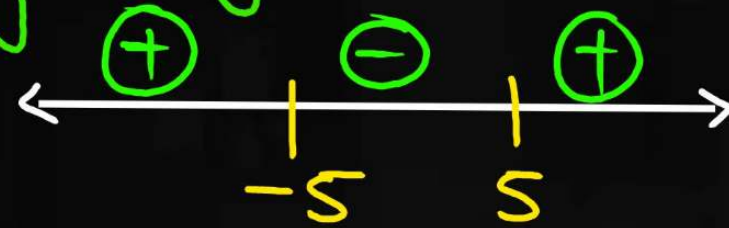
$$\Rightarrow -(25 - x^2) \leq 0$$

$$\Rightarrow x^2 - 25 \leq 0$$

$$\Rightarrow x^2 - 5^2 \leq 0$$

$$\Rightarrow (x + 5)(x - 5) \leq 0$$

So By wavy - Curve Method (Sign-scheme method)



$$\Rightarrow x \in [-5, 5]$$

Therefore Domain =  $[-5, 5]$ .

Further for  $x \in [-5, 5]$ , we have  $x^2 \geq 0$

$$\& \ x^2 \leq 25.$$

i.e

$$0 \leq x^2 \leq 25$$



$$\Rightarrow -25 \leq -x^2 \leq 0$$

$$\Rightarrow \cancel{-25} + \cancel{25} \leq 25 - x^2 \leq 0 + 25$$

$$\Rightarrow 0 \leq 25 - x^2 \leq 25$$

$$\Rightarrow 0 \leq \sqrt{25 - x^2} \leq \sqrt{25}$$

$$\Rightarrow 0 \leq f(x) \leq 5$$

Therefore Range =  $[0, 5]$

$$* -5 \leq x \leq 5$$

$$0 \leq x^2 \leq 25$$

$$\Rightarrow -0 \geq -x^2 \geq -25$$

$$\Rightarrow -25 \leq -x^2 \leq 0$$

$$\Rightarrow \cancel{-25} + \cancel{25} \leq 25 - x^2 \leq 0 + 25$$

$$\Rightarrow 0 \leq 25 - x^2 \leq 25$$

$$\Rightarrow \sqrt{0} \leq \sqrt{25 - x^2} \leq \sqrt{25}$$

$$\Rightarrow 0 \leq f(x) \leq 5$$

## Question

Domain of  $\sqrt{a^2 - x^2}$  ( $a > 0$ ) is

~~1~~  $(-a, a)$

☒ 2  $[-a, a]$

~~3~~  $[0, a]$

~~4~~  $(-a, 0)$

(Sol) Given function is,

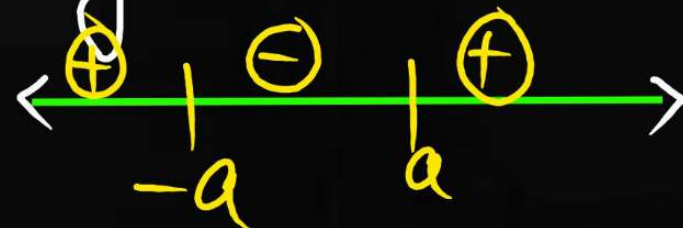
$$f(x) = \sqrt{a^2 - x^2}$$

We know that  $f(x)$  is defined only if  $a^2 - x^2 \geq 0$

$$\Rightarrow x^2 - a^2 \leq 0$$

$$\Rightarrow (x+a)(x-a) \leq 0$$

By wavy - Curve Method,



$$\Rightarrow x \in [-a, a]$$

So, Domain =  $[-a, a]$ .

## Question

Domain of  $\sqrt{x^2 - a^2}$  ( $a > 0$ ) is

- ☒ 1  $(-a, a)$
- ☒ 2  $[-a, a]$
- ☒ 3  $[0, a]$
- ☒ 4  $(-\infty, a] \cup [a, \infty)$

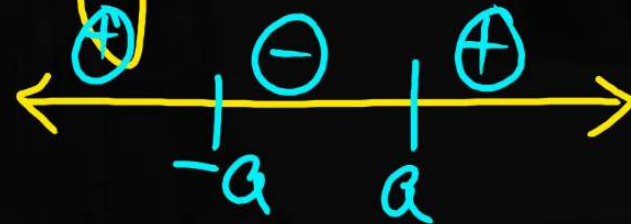
(Sol) Given function is,

$$f(x) = \sqrt{x^2 - a^2} \quad (a > 0)$$

We know that  $f(x)$  is defined only if  $x^2 - a^2 \geq 0$

$$\Rightarrow (x+a)(x-a) \geq 0$$

By wavy-Curve Method



$$\Rightarrow x \in (-\infty, -a] \cup [a, \infty)$$

$$\text{So Domain} = (-\infty, -a] \cup [a, \infty)$$



## Question

The domain of the function  $f$  given by  $f(x) = \frac{x^2 + 2x + 1}{x^2 - x - 6}$  (Rational function)

☒ 1  $\mathbb{R} - \{3, -2\}$

☐ 2  $\mathbb{R} - \{-3, 2\}$

☐ 3  $\mathbb{R} - [3, -2]$

☐ 4  $\mathbb{R} - (3, -2)$

(Sol) Here  $x^2 - x - 6 = 0$

$$\Rightarrow x^2 - 3x + 2x - 6 = 0$$

$$\Rightarrow x(x-3) + 2(x-3) = 0$$

$$\Rightarrow (x-3)(x+2) = 0$$

$$\Rightarrow x = 3, -2$$

So, Domain =  $\mathbb{R} - \{3, -2\}$

## Question

The domain and range of the real function  $f$  defined by  $f(x) = \frac{4-x}{x-4}$  is given by

- ☒ 1 Domain =  $\mathbf{R}$ , Range =  $\{-1, 1\}$
- ☒ 2 Domain =  $\mathbf{R} - \{1\}$ , Range =  $\mathbf{R}$
- ☒ 3 Domain =  $\mathbf{R} - \{4\}$ , Range =  $\{-1\}$
- ☒ 4 Domain =  $\mathbf{R} - \{-4\}$ , Range =  $\{-1, 1\}$

$$x - 4 = 0$$

$$x = 4$$

$$\text{So Domain} = \mathbf{R} - \{4\}$$

$$x \neq 4$$

$$f(x) = \frac{4-x}{x-4} = \frac{-(x-4)}{x-4}$$

$$\Rightarrow f(x) = -1 \quad \forall x \neq 4 \text{ and } x \in \mathbf{R}$$

$$\text{Range} = \{-1\}$$



## Question

The domain and range of real function  $f$  defined by  $f(x) = \sqrt{x-1}$  is given by

☒ 1 Domain =  $(1, \infty)$ , Range =  $(0, \infty)$

☒ 2 Domain =  $[1, \infty)$ , Range =  $(0, \infty)$

☒ 3 Domain =  $[1, \infty)$ , Range =  $[0, \infty)$

☒ 4 None of these

$$\begin{aligned} * x-1 &\geq 0 \\ x &\geq 1 \end{aligned}$$

$$\begin{aligned} \sqrt{x-1} &\geq \sqrt{0} \\ \sqrt{x-1} &\geq 0 \\ f(x) &\geq 0 \end{aligned}$$

$$\text{Domain} = [1, \infty)$$

$$\text{Range} = [0, \infty)$$

## Question

$$x < -1 \text{ or } x > 1$$

The domain of the function  $f$  defined by  $f(x) = \sqrt{4-x} + \frac{1}{\sqrt{x^2-1}}$  is equal to

☒ 1  $(-\infty, -1) \cup [1, 4]$

☒ 2  $(-\infty, -1] \cup [1, 4]$

☒ 3  $(-\infty, -1) \cup [1, 4]$

☒ 4  $(-\infty, -1) \cup [1, 4]$

(Sol) We know that  $f(x) = \sqrt{4-x} + \frac{1}{\sqrt{x^2-1}}$

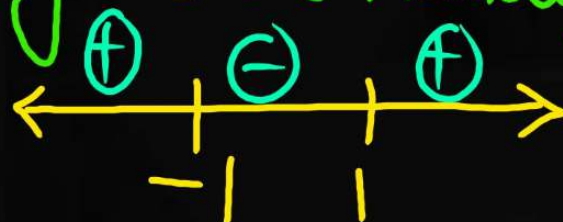
is defined only if

$$4-x \geq 0 \text{ and } x^2-1 > 0$$

$$\Rightarrow -x \geq -4 \text{ \& } x^2-1^2 > 0$$

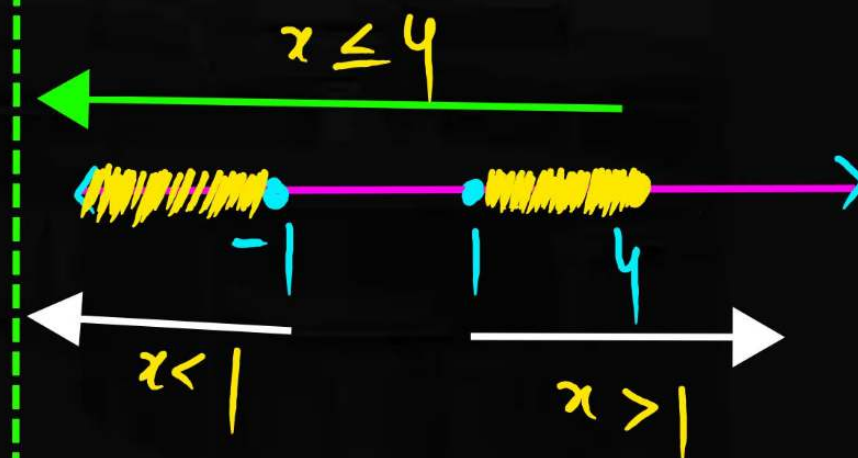
$$\Rightarrow x \leq 4 \text{ and } (x+1)(x-1) > 0$$

By wavy-Curve Method, we get;



$$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$$

$$\text{So Domain} = (-\infty, -1) \cup [1, 4]$$





## Question

The domain for which the functions defined by  $f(x) = 3x^2 - 1$  and  $g(x) = 3 + x$  are equal to

☒ 1  $\left\{-1, \frac{4}{3}\right\}$

☐ 2  $\left[-1, \frac{4}{3}\right]$

☐ 3  $\left(-1, \frac{4}{3}\right)$

☐ 4  $\left[-1, \frac{4}{3}\right)$

$$(Sol) f(x) = g(x)$$

$$\Rightarrow 3x^2 - 1 = 3 + x$$

$$\Rightarrow 3x^2 - x - 4 = 0$$

$$\Rightarrow 3x^2 - 4x + 3x - 4 = 0$$

$$\Rightarrow x(3x - 4) + 1(3x - 4) = 0$$

$$\Rightarrow (x + 1)(3x - 4) = 0$$

$$\Rightarrow x = -1, \frac{4}{3}$$

## Question

If  $f(x) = ax + b$ , where  $a$  and  $b$  are integers,  $f(-1) = -5$  and  $f(3) = 3$ , then  $a$  and  $b$  are equal to

**1**  $a = -3, b = -1$

**2**  $a = 2, b = -3$

**3**  $a = 0, b = 2$

**4**  $a = 2, b = 3$

Homework  
Problem.



**Example**

Find the domain for which the functions  $f(x) = 2x^2 - 1$  and  $g(x) = 1 - 3x$  are equal.

$$f(x) = g(x)$$

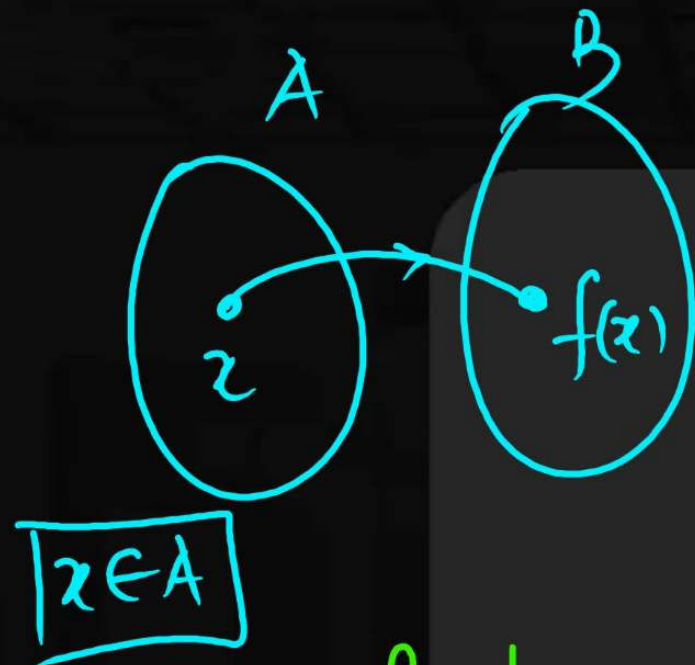
Home work  
Problem.

\*\*\*\*\*

LIBA  
HAMID  $\rightarrow$  OP

# Thank You

$\frac{1}{0}$



$$R = \{(x, f(x)) : x \in A\}$$

$$f = \{(x, f(x)) : x \in A, f(x) \in B\}$$